Technology Stack

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence (CONSTRUCTAI)	**Maximum Marks:**	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details (Part 1 - Client, Server & Storage)

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** **(UI & Real-time Visuals)**	**React.js** (Vite) Tailwind CSS Lucide React Icons	Enables a modular, responsive single-page application (SPA). Provides high-contrast UI components for outdoor daylight visibility, drag-and-drop image upload, and real-time bounding box inspection views.
2	**Backend / Server-Side** **(API & Logic Layer)**	**FastAPI** (Python 3.10+) Uvicorn ASGI Server Pydantic	Asynchronous high-throughput framework with native OpenAPI documentation. Delivers sub-millisecond route handling, strict request validation, and seamless non-blocking I/O for heavy vision payloads.
3	**Database / Data Storage** **(Telemetry & Audits)**	**SQLite** / Local Structured JSON Store & In-Memory Cache	Zero-configuration, lightweight local transactional persistence. Ideal for local test deployments, audit trail persistence, BOQ item catalogs, and cached IS 456 remediation guidelines without external database overhead.

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Technology Stack Details (Part 2 - Execution, Version Control & APIs)

Detailing execution scripts, container environment, source control, and multimodal GenAI integrations.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	**Cloud / Hosting / Deployment** **(Execution Environment)**	**Local Execution** via Windows Batch Scripts (run_backend.bat, run_frontend.bat) & Docker Ready	Configured for rapid one-click local setup and live demonstration without relying on recurring cloud costs or internet latency during on-site evaluations. Container-ready for seamless cloud deployment when required.
5	**Version Control & CI/CD** **(Code Management)**	**Git & GitHub** (Repository Structured to NASSCOM Standards)	Tracks iterative feature updates, maintains structured branch history, isolates environment variables via .gitignore, and allows drag-and-drop manual synchronization directly into NASSCOM capstone portal.
6	**Third-Party APIs / Other Tools** **(AI & Weather Services)**	**Groq AI Cloud API** (LLaMA 3.2 11B/90B Vision) Open-Meteo Weather API ReportLab PDF Engine	• **Groq LLaMA 3.2 Vision:** Sub-second multi-modal vision reasoning for PPE violation spotting & IS 456 concrete crack classification. • **Open-Meteo:** Live rainfall & wind speed threshold monitoring. • **ReportLab:** Automated executive PDF audit report generation.